

Lab: Specific Heat Capacity of Aluminum & Leakiness Test

Aaron Huang

March 25, 2026

I Purpose

To find the specific heat capacity of aluminum by experiment & compare the experimental value to reference material

II Introduction

This lab was designed to determine the specific heat capacity of aluminum by experiment using calorimetry. The specific heat capacity of a substance is the amount of energy required to raise the temperature of one gram of the substance by one degree Celsius. This experiment, used a calorimeter, a device that minimizes heat and mass transfer between the system and its surroundings, to measure the heat transfer between aluminum and water. Additionally, the experiment included a leakiness test to determine the rate at which thermal energy leaks out of the calorimeter, which is important for accurate measurements.

The procedure involved heating aluminum pellets in a hot water bath until they reached the temperature of boiling water, 100°C , then the aluminum was transferred to a calorimeter containing water with known mass and initial temperature. The maximum temperature reached by the water was recorded, as well as the time taken to reach that temperature. Finally, after waiting for 3 minutes, the temperature of the water once again was recorded to determine the rate of heat loss from the calorimeter.

The hot plate, boiling water, and aluminum pellets were handled with care to avoid burns and the rubber stopper was put on loosely to prevent pressure buildup. Additionally, the hot plate was not moved while in use and unplugged after use to prevent accidents.

III Observations and Results

Table 1: Experimental Data for Specific Heat of Aluminum

Measurement	Unit	Trial 1	Trial 2	Trial 3
Mass of empty calorimeter	g			
Mass of calorimeter + water	g			
Mass of aluminum pellets (m_{Al})	g			
Initial temp of water ($T_{i,w}$)	$^{\circ}\text{C}$			
Initial temp of Al ($T_{i,Al}$)	$^{\circ}\text{C}$			
Maximum temp reached (T_{max})	$^{\circ}\text{C}$			
Time to reach T_{max} (Δt)	s		—	—
Temp after 3-minute wait (T_{3min})	$^{\circ}\text{C}$		—	—